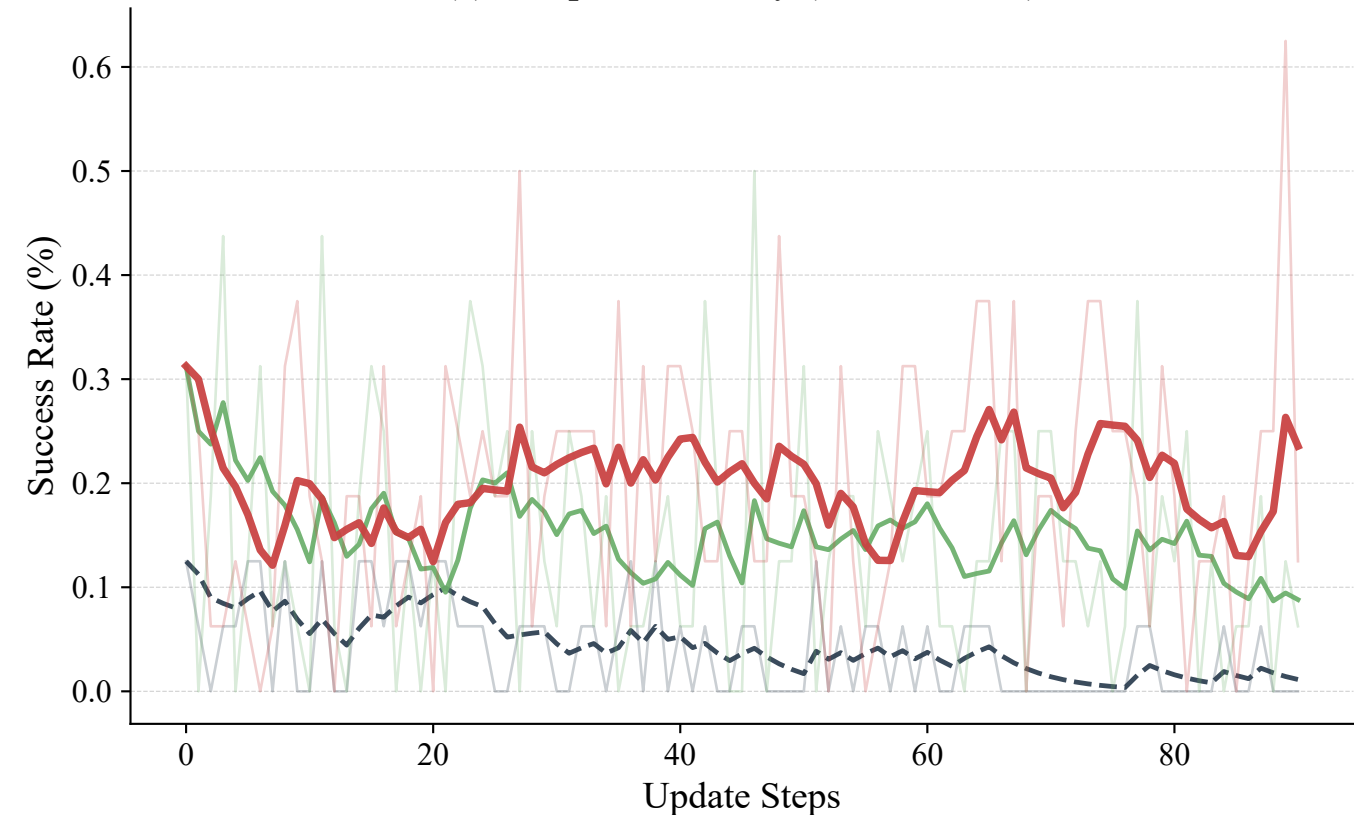


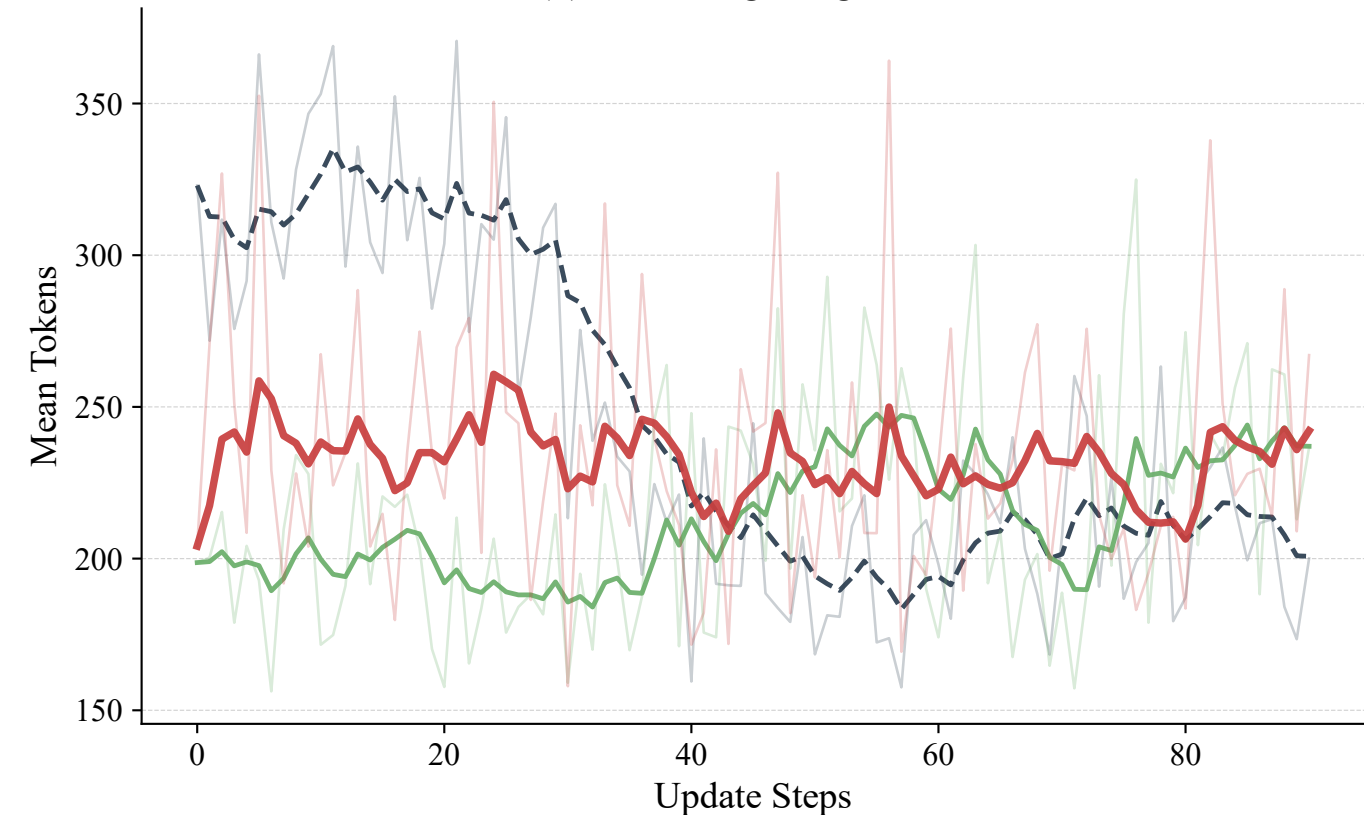
PPO vs. GRPO Scaling & Efficiency Analysis (Arithmetic-24)



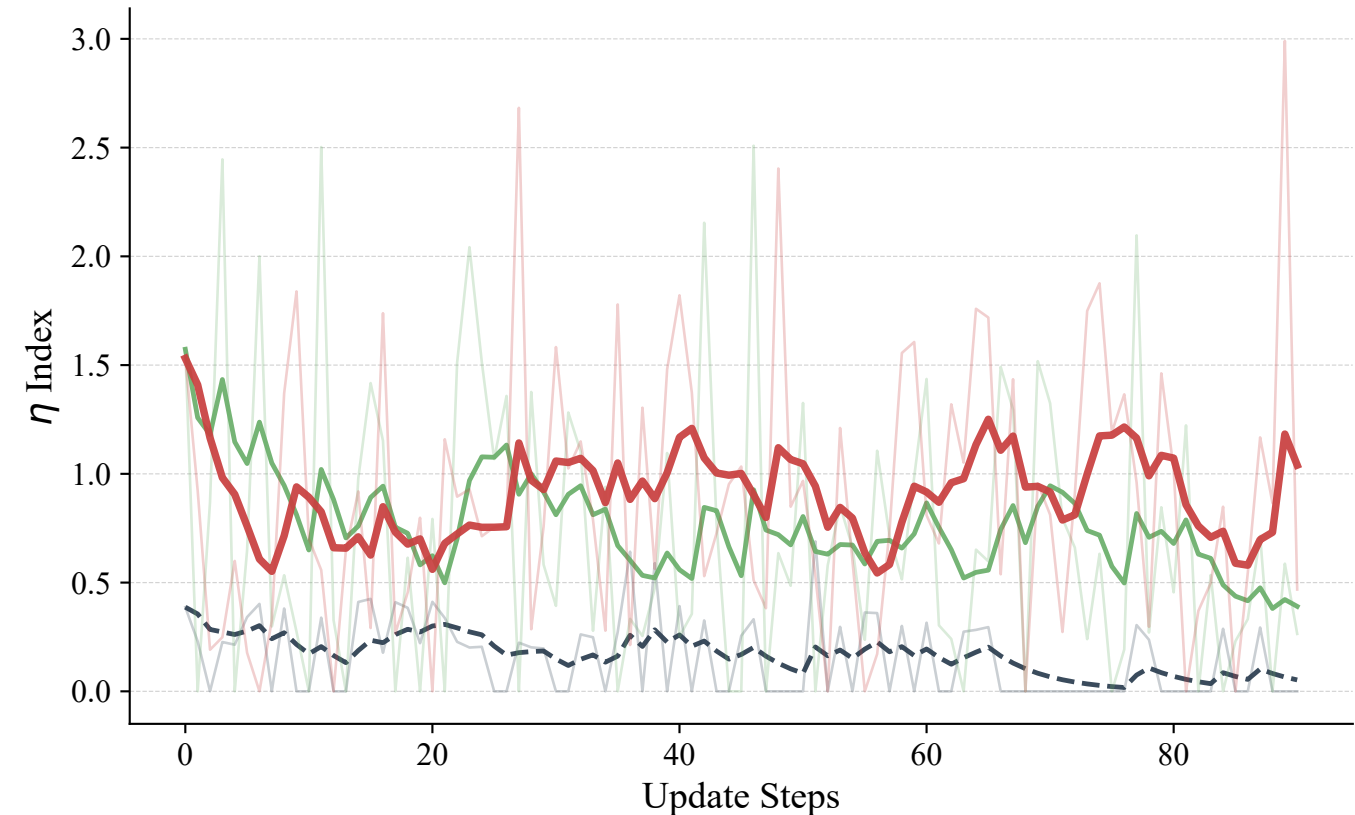
(a) Sample Efficiency (Success Rate)



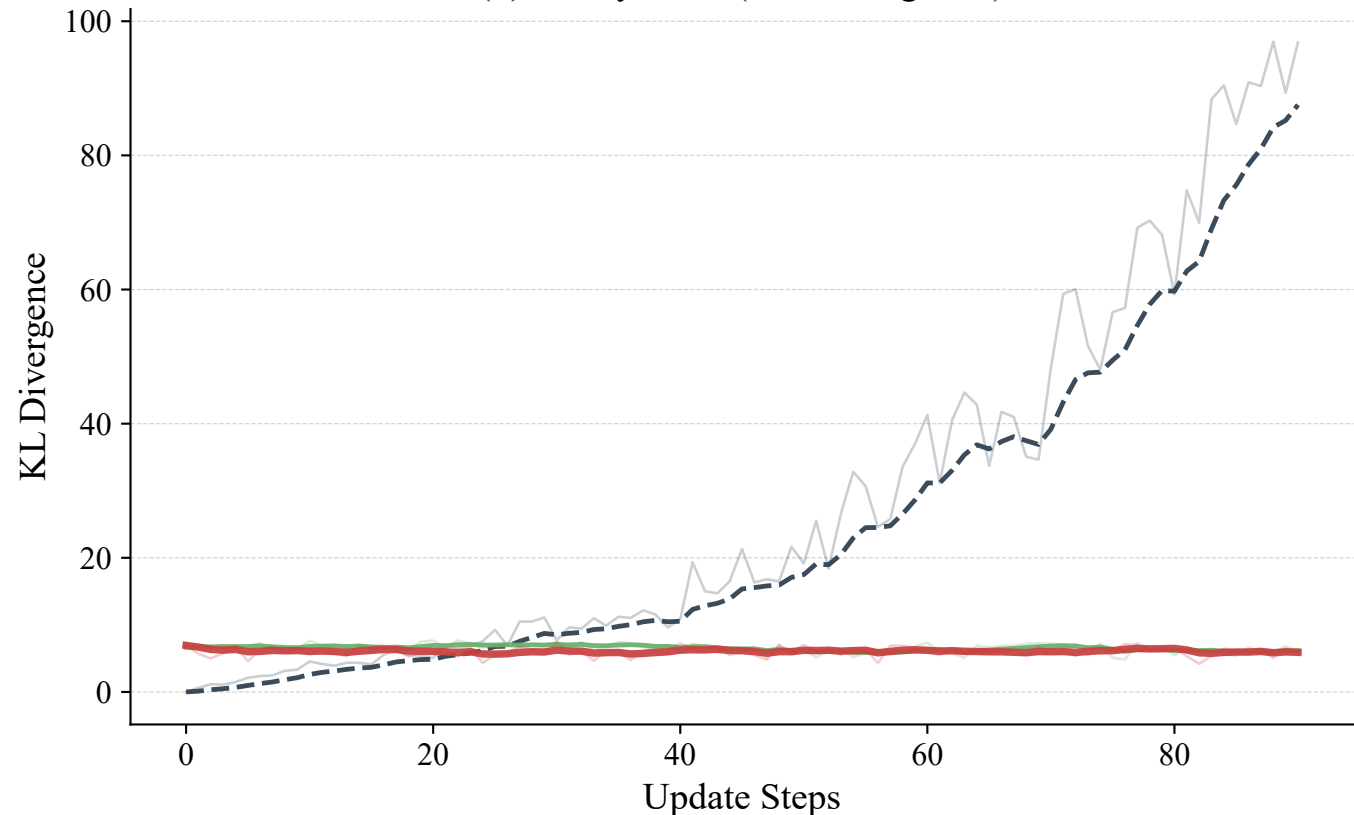
(b) Reasoning Length Cost



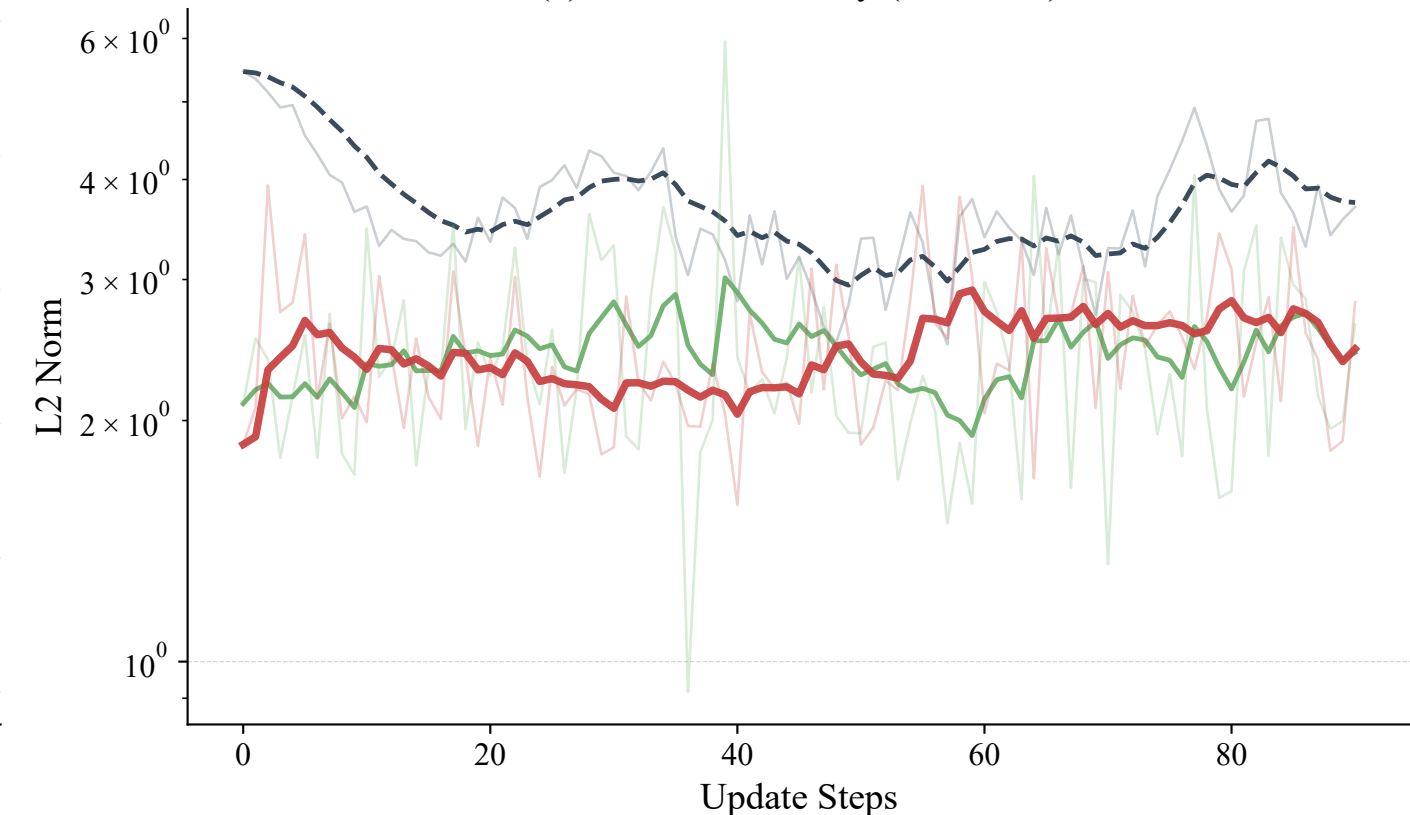
(c) Efficiency Index (η)



(d) Policy Drift (KL Divergence)



(e) Gradient Stability (L2 Norm)



(f) Gradient Variance

